Depu Meng, Ph. D.

Senior Research Engineer and Tech Lead at Applied Intuition

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Research Interests

Multimodal intelligence: Multimodal pretraining, image/video understanding and generation, spatial-temporal reasoning, world models, large-scale training and data systems

Education

Microsoft Research Asia - University of Science and Technology of China — Beijing, China

Ph. D. in Control Science and Engineering — *Sept. '18 - Jun. '23*

- Advisors: Dr. Baining Guo (Microsoft), Prof. Houqiang Li (USTC)

University of Science and Technology of China — Hefei, China

B.E. in Electrical Engineering (School of Gifted Young) — *Sept. '14 - Jun. '18*

Work Experience

Applied Intuition, Research Team — Sunnyvale, CA

Senior Research Engineer and Tech Lead — *Sep. '25 - present*

- **Multimodal Foundation Model - Scalable Training.** Built the 0-to-1 framework for 5B-20B world models with heterogeneous sensor/action I/O, scaling to 500+ GPUs via DDP and ZeRO 1-3.
- Lead training, encoder, throughput, GPU optimization, and heterogeneous loading; co-design ingestion, filtering, curation, LanceDB, autolabeling, synthetic data, and external acquisition.
- **World-Model Research - Multimodal Reasoning.** Advance spatial-temporal memory and reasoning, physical realism in video generation, and 3D reconstruction and generation.
- Develop multi-step reasoning across text, image, and video; as Tech Lead for four FTEs and four interns, set direction, coordinate resources, mentor researchers, and develop papers.
- **End-to-End Driving - Training & Evaluation.** As one of the first two modeling engineers, rebuilt the training platform, initiated data curation and a road-test flywheel, and led pretraining stability and quality evaluation.

DiDi Autonomous Driving, AI Research Team — San Jose, CA

Machine Learning Engineer — *Dec. '24 - Aug. '25*

- Built a 3D Gaussian Splatting framework for camera simulation, improving rendering through distortion handling and an in-house diffusion-based refinement model.
- Productionized the simulator with approximately 10x throughput via GPU-memory caching, fewer CPU-GPU transfers, and batched multi-sensor rendering.
- Developed end-to-end model pretraining and road-test-driven interaction and safety capabilities for vehicles and vulnerable road users.

University of Michigan — Ann Arbor, MI

Post-Doctoral Research Fellow — *Aug. '23 - Dec. '24*

- Researched safety-critical event identification, multimodal perception, sensor fusion, and automated data synthesis for roadside and connected-vehicle systems.
- Managed four postdoctoral/Ph.D. researchers for the U.S. DOT Intersection Safety Challenge and owned object-detection quality; the team won Stages 1A and 1B.

Internship Experience

University of Michigan — Ann Arbor, MI
Research Assistant — *Apr. '22 - Aug. '23*

Meituan Autonomous Delivery Group — Beijing, China
Research Intern — *Aug. '21 - Apr. '22*

Microsoft Research Asia, Visual Computing Group — Beijing, China
Research Intern — *Jul. '19 - Jul. '21; Jul. '17 - Jul. '18*

Publications

⋆ *equal contribution,* † *project lead.*

STRIKE: Learning Visual State Transitions for Physical World Modeling
Wenbin Teng, Tianshuo Xu, **Depu Meng**†, Yuelei Li, Quentin Herau, Yihan Hu, Yajie Zhao, Wei Zhan
arXiv preprint, 2026 [PAPER] [PROJECT]

S2Tok: Streaming 3D Gaussian Reconstruction with Persistent Spatial Tokens
Fang Li, Jiraphon Yenphraphai, Quentin Herau, **Depu Meng**, Yihan Hu, Tianshuo Xu, Narendra Ahuja, Wei Zhan
arXiv preprint, 2026 [PAPER] [PROJECT]

Building Rome from a Single Image
Jiraphon Yenphraphai, Fang Li, Tianshuo Xu, **Depu Meng**, Quentin Herau, Yihan Hu, Raymond A. Yeh, Wei Zhan
arXiv preprint, 2026 [PAPER] [PROJECT]

Teaching Video Generators to Remember: Eliciting Dynamic Memory for Out-of-Sight State Evolution
Tianshuo Xu, Yichen Xie, **Depu Meng**†, Chensheng Peng, Quentin Herau, Bo Jiang, Yihan Hu, Wei Zhan
NeurIPS, 2026 [PAPER] [PROJECT] [CODE] [MODEL] [DATASET]

LaMo: Self-Supervised Latent Motion Priors for Physical Realism in Video Generation
Bo Jiang, **Depu Meng**, Yihan Hu, Yichen Xie, Tianshuo Xu, Wei Zhan
NeurIPS, 2026 [PAPER] [PROJECT] [CODE]

SpectralSplat: Appearance-Disentangled Feed-Forward Gaussian Splatting for Driving Scenes
Quentin Herau⋆, Tianshuo Xu⋆, **Depu Meng**†, Jiezhi Yang,
Chensheng Peng, Spencer Sherk, Yihan Hu, Wei Zhan
arXiv preprint, 2026 [PAPER]

URoPE: Universal Relative Position Embedding across Geometric Spaces
Yichen Xie, **Depu Meng**, Chensheng Peng, Yihan Hu, Quentin Herau, Masayoshi Tomizuka, Wei Zhan
ECCV, 2026 (*Spotlight*) [PAPER] [PROJECT] [CODE]

Exploring Communication and Roadside Perception Requirements for Cooperative Warning Systems at Intersections
Tinghan Wang, **Depu Meng**, Boqi Li, Rusheng Zhang, Yukun Zuo, Shengyin Shen, Darian Hogue, Michael Maile, Michael Shulman, Henry X. Liu
IEEE IV, 2025 [PAPER]

Towards Comprehensive Roadside Intelligence: Sensor Fusion and Full-Stack Perception with Multiple Cameras
Rusheng Zhang, **Depu Meng**, Shengyin Shen, Boqi Li, Tinghan Wang, Henry X. Liu
IEEE IV, 2025 [PAPER]

MSight: An Edge-Cloud Infrastructure-Based Perception System for Connected Automated Vehicles
Rusheng Zhang*, **Depu Meng***, Shengyin Shen, Zhengxia Zou, Houqiang Li, Henry X. Liu
Technical Report, 2024 [PAPER] [CODE]

Robust Roadside Perception: An Automated Data Synthesis Pipeline Minimizing Human Annotation
Rusheng Zhang, **Depu Meng**, Lance Bassett, Shengyin Shen, Zhengxia Zou, Henry X. Liu
IEEE T-IV, 2024 [PAPER]

Evaluating Roadside Perception for Autonomous Vehicles: Insights from Field Testing
Rusheng Zhang, **Depu Meng**, Shengyin Shen, Tinghan Wang, Tai Karir, Michael Maile, Henry X. Liu
arXiv preprint, 2024 [PAPER]

ROCO: A Roundabout Traffic Conflict Dataset
Depu Meng, Owen Sayer, Rusheng Zhang, Shengyin Shen, Houqiang Li, Henry X. Liu
TRB Annual Meeting, 2023 [PAPER] [DATASET]

HyMo: Hybrid Motion Representation Learning for Prediction from Raw Sensor Data
Depu Meng, Changqian Yu, Jiajun Deng, Deheng Qian, Houqiang Li, Dongchun Ren
IEEE T-MM, 2023 [PAPER]

CORE: Consistent Representation Learning for Face Forgery Detection
Yunsheng Ni, **Depu Meng**, Changqian Yu, Chengbin Quan, Dongchun Ren, Youjian Zhao
CVPR Workshops, 2022 [PAPER] [CODE]

Conditional DETR for Fast Training Convergence
Depu Meng*, Xiaokang Chen*, Zejia Fan, Yuhui Yuan, Gang Zeng, Houqiang Li, Lei Sun, Jingdong Wang
ICCV, 2021 [PAPER] [CODE] [MODEL]

Consistent Instance Classification for Unsupervised Representation Learning
Depu Meng, Zigang Geng, Zhirong Wu, Bin Xiao, Houqiang Li, Jingdong Wang
ICCV Workshops, 2021 [PAPER]

Bottom-Up Human Pose Estimation by Ranking Heatmap-Guided Adaptive Keypoint Estimates
Ke Sun, Zigang Geng, **Depu Meng**, Bin Xiao, Dong Liu, Zhaoxiang Zhang, Jingdong Wang
arXiv preprint, 2020 [PAPER] [CODE]

Deep Convolutional Neural Networks with Merge-and-Run Mappings
Liming Zhao, Mingjie Li, **Depu Meng**, Xi Li, Zhuowen Tu, Zhaoxiang Zhang, Yueting Zhuang, Jingdong Wang
IJCAI, 2018 [PAPER] [CODE]

Awards

U.S. DOT Intersection Safety Challenge Stage 1B Tier 1 Winner Team (Prize: $750,000) *Jan. '25*

U.S. DOT Intersection Safety Challenge Stage 1A Winner Team (Prize: $100,000) *Jan. '24*

Services

Conference Reviewer: CVPR, ECCV, ICCV, NeurIPS, ACM MM, TRBAM, IROS

Journal Reviewer: IJCV, IEEE T-IV, IEEE T-MM, IEEE T-CSVT, Neurocomputing, Pattern Recognition